

Climate 5000 Series - Gen 4

9K BTU Single Zone Regular System

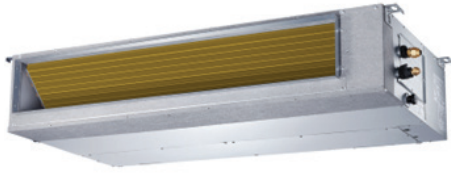
Ducted IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

9K Btu Single Zone Regular System - Ducted IDU

Indoor part number		8-733-962-709
Indoor model number		BMS500-AAU009-1AHDXD
Outdoor part number		8-733-962-684
Outdoor model number		BMS500-AAS009-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		19.5
EER2		12.8
HSPF2 Region IV		11.0
HSPF2 Region V		8.8
COP(47F)		3.66
COP (5F)		2.25
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3100~11900
	Rated Capacity	9000
	Current	4.7
	Input Power	703
Heating	Capacity Range	3600~14000
	Rated Capacity (47F)	10000
	Rated Capacity (5F)	8600
	Current	3.6
	Input Power	800
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 12.0
Max. fuse indoor/outdoor (A)		15.0 / 15.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 310
Input (W)		325 / 826
Rated current (RLA/A)		2.40 / 5.65
Piping		
Refrigerant precharge (oz/kg)		32.45 / 0.92
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		82.00 / 25
Max. difference in height (ft/m)		49.20 / 15

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Operation Range

ODU	Cooling (°F / °C)	-13 ~ 122 / -25 ~ 50
	Heating (°F / °C)	-13 ~ 75 / -25 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

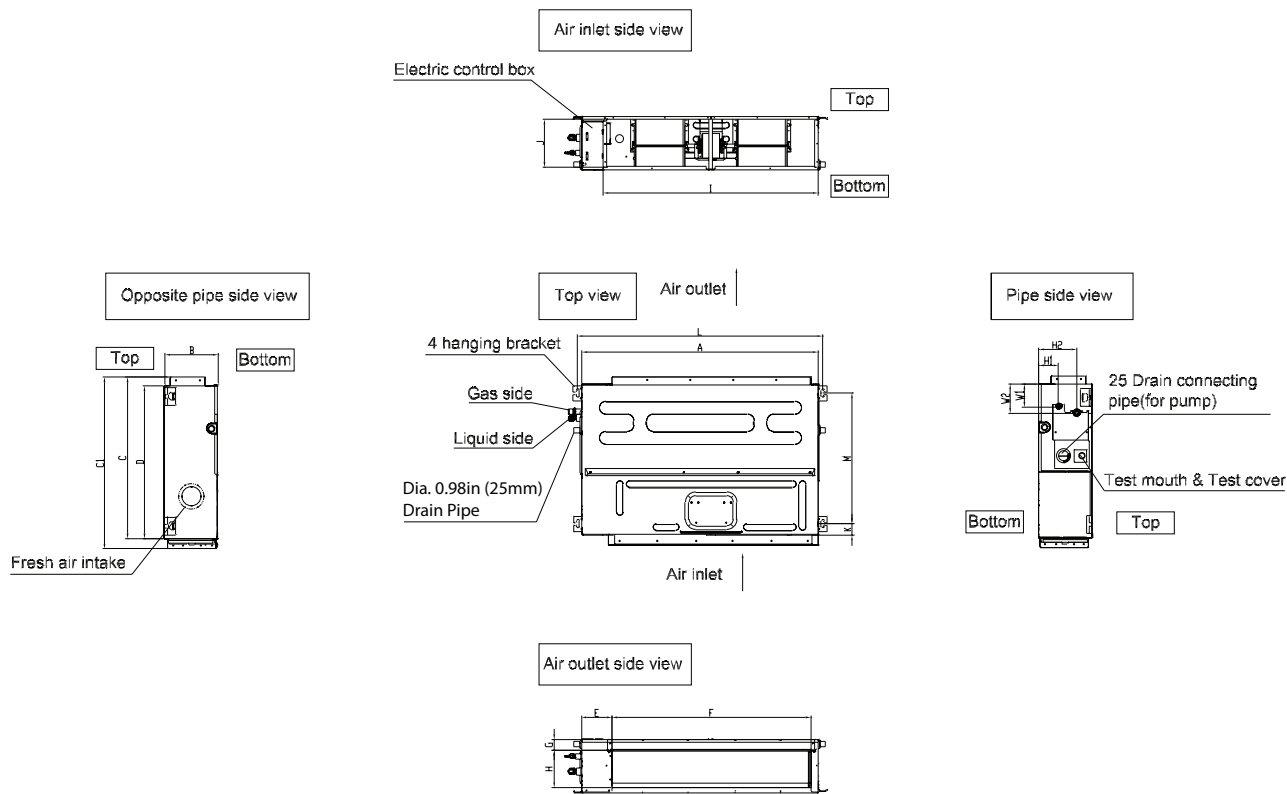
System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		1.5
IDU	Fan motor model	ZKFN-81-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	150.0
	Fan speed, H / M / L (RPM)	980 / 910 / 840
	Air flow, Turbo / H / M / L / Si (CFM)	323.7 / 264.9 / 206.0 / 147.2 / 117.7
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 34 / 32.5 / 29 / 24
	Net weight (lbs / kg)	39.46 / 17.9
	Gross weight (lbs / kg)	46.74 / 21.2
ODU	ESP range (Pa)	0-100
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-34-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed, H / / L (RPM)	800 / 700 / 600
	Air flow, Max (CFM)	1236.1
	Noise level, Max (dB(A))	54.0
Accessories	BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
	BMS-DRY-XXD (8733969263)	24V Interface
Accessories	BMS-PK1-7X1 (8733951010)	25ft Lineset

Bosch Thermotechnology Corp.
Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

Tel: 1-800-283-3787 www.bosch-homecomfort.us

Indoor Unit Dimensions and Connections

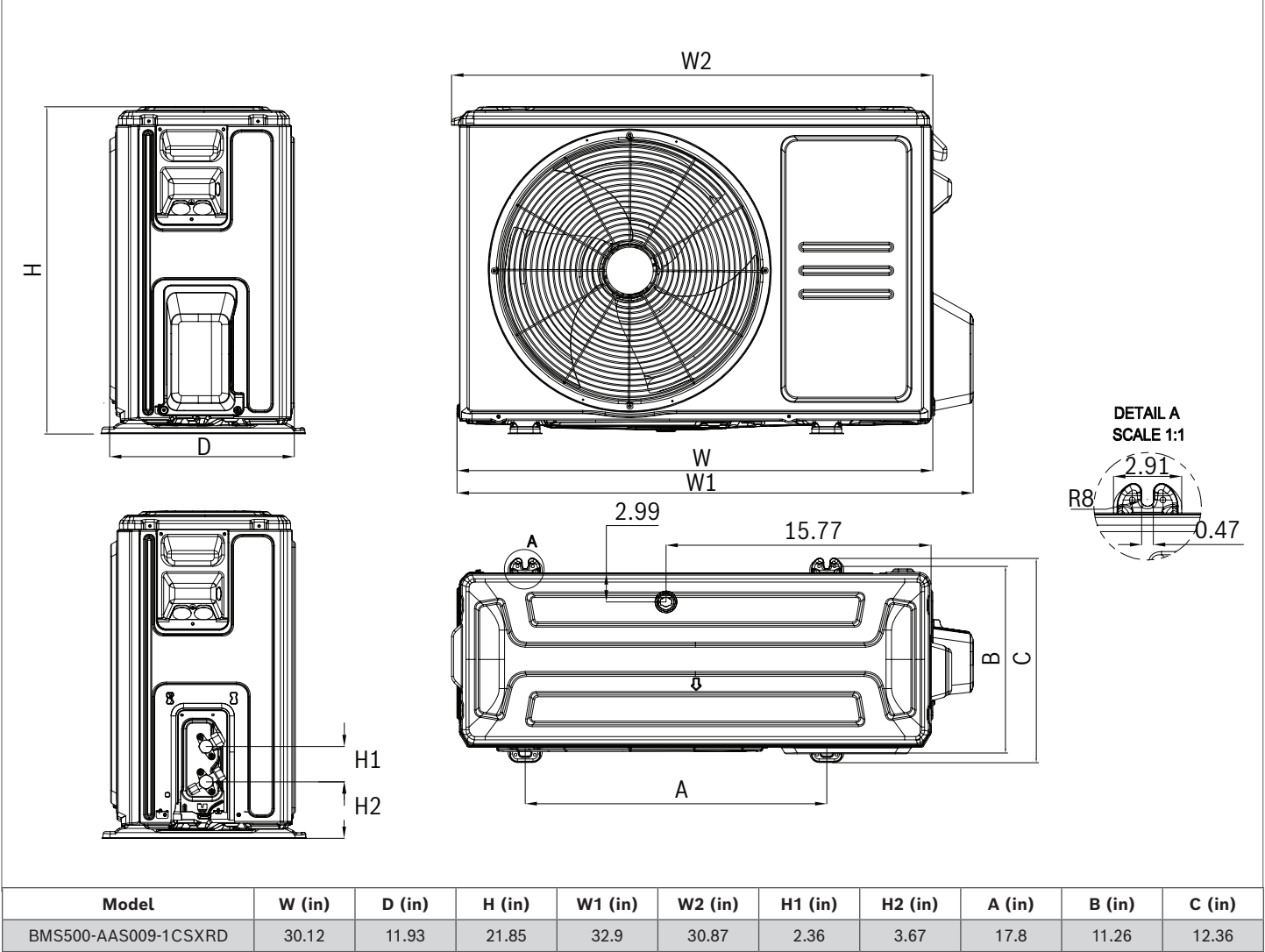


Model	Outline dimension (in)				Air outlet dimension (in)				Return air (in)			Mounting (in)		Gas and liquid valve (in)			
	A	B	C	D	E	F	G	H	I	J	K	L	M	H1	H2	W1	W2
BMS500-AAU009-1AHDXD	27.6	7.9	19.9	17.7	5.4	21.1	1.2	6.0	23.6	7.3	2.0	29.2	14.2	3.3	5.5	3.3	3.3

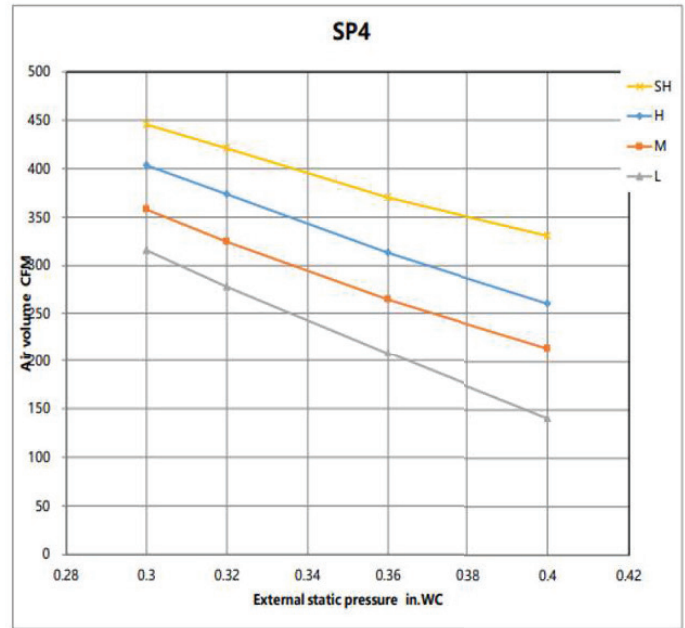
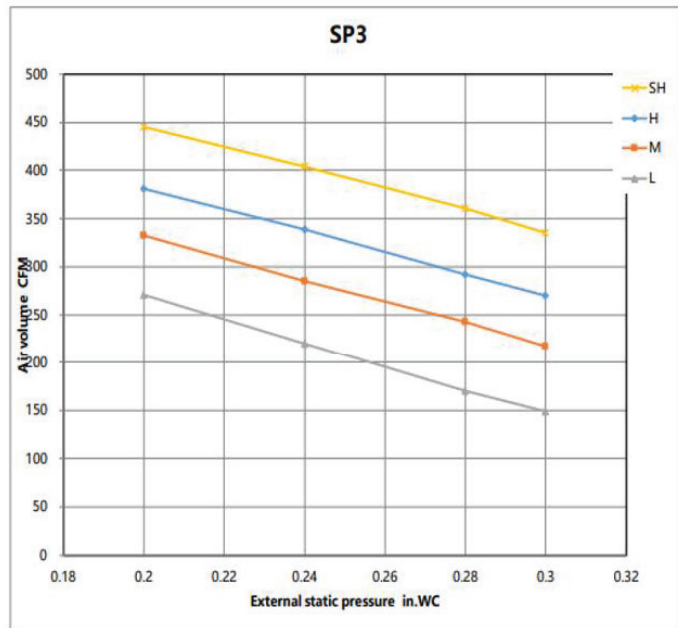
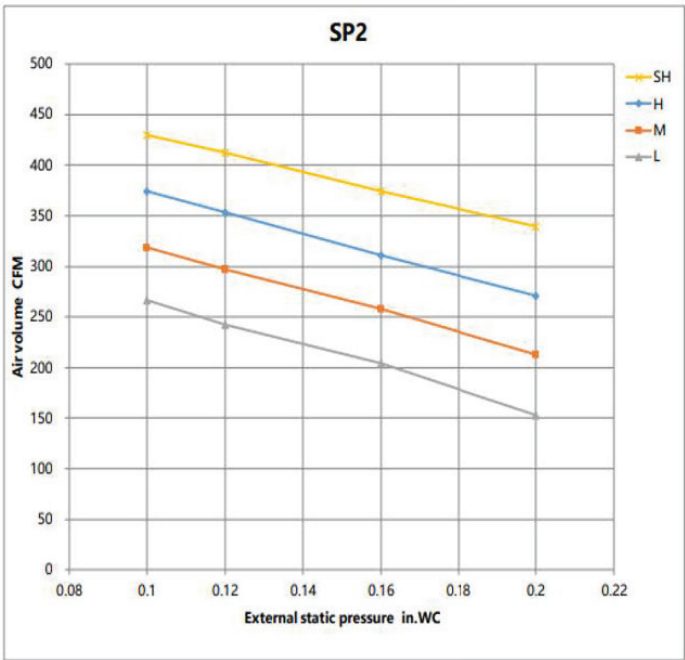
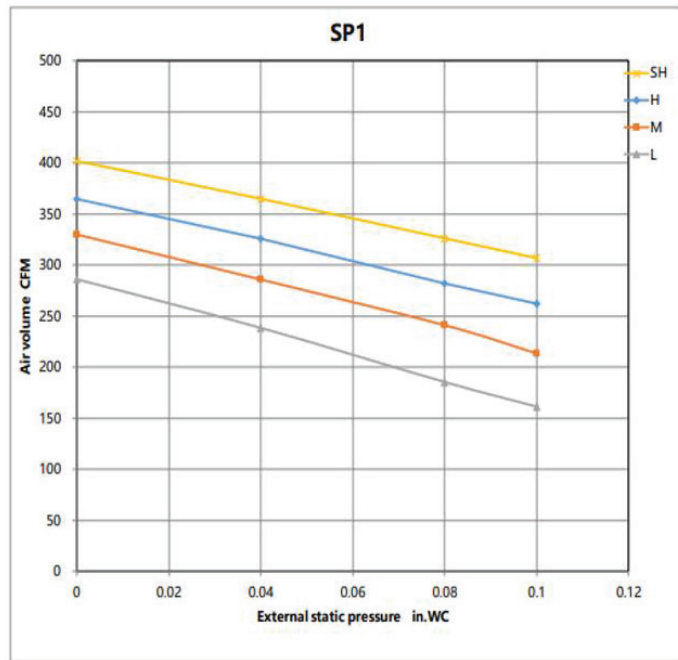
Climate 5000 Series - Gen 4
9K Btu Single Zone Regular System - Ducted IDU



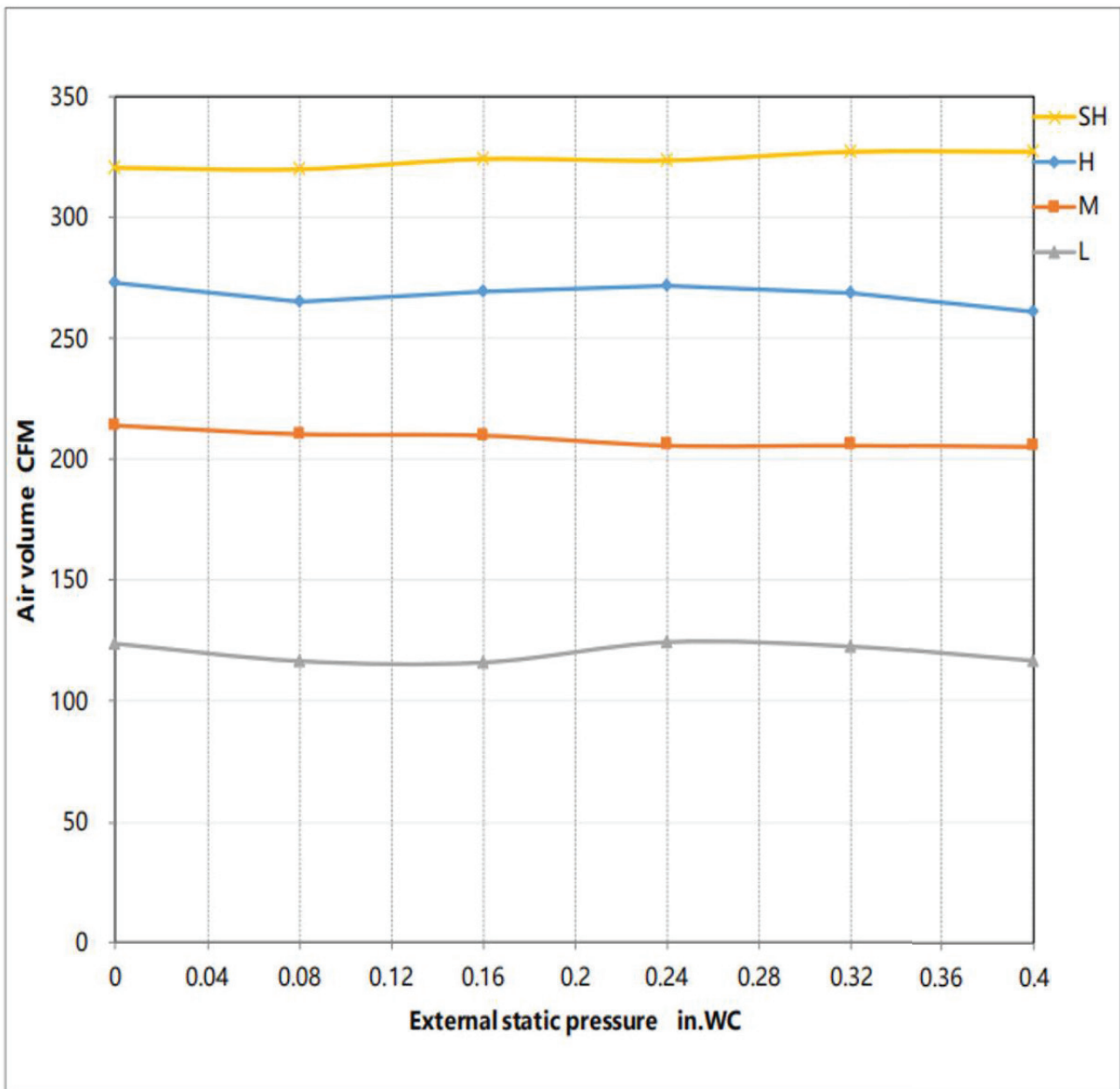
Outdoor Unit Dimensions and Connections



Air Volume



Air Volume



9K Btu Single Zone Regular System - Ducted IDU



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Cooling Performance - Indoor Dry Bulb Temperature at 324 CFM																					
ODU DB(F)	IDU WB (F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	7463	7460	7566	7597	7807	7787	7780	7775	7768	8019	8009	8002	8001	7993	8459	8457	8439	8436	8418	
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.52	0.52	0.52	0.52	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.52	0.52	0.52	0.52	0.52
-4°F (-20.0°C)	TC	8037	8034	8148	8182	8407	8386	8379	8373	8373	8366	8636	8626	8617	8616	8608	9110	9107	9089	9085	9066
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.53	0.54	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53
0°F (-17.8°C)	TC	8304	8299	8453	8488	8721	8660	8651	8644	8643	8730	8915	8902	8891	8890	8885	9394	9392	9391	9392	9398
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.55	0.55	0.55	0.55	0.55
5°F (-15.0°C)	TC	8542	8571	8767	8808	9079	8943	8945	8945	8955	9088	9229	9231	9233	9233	9245	9770	9767	9772	9773	9777
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.54	0.54	0.54	0.54	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.56	0.56	0.56	0.56	0.56
14°F (-10.0°C)	TC	9181	9176	9364	9408	9699	9637	9628	9622	9621	9711	9967	9954	9944	9943	9933	10598	10595	10574	10571	10548
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61
17°F (-8.3°C)	TC	9371	9402	9619	9666	9980	9838	9842	9844	9852	9991	10174	10177	10181	10182	10194	10820	10817	10815	10817	10826
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63
23°F (-5.0°C)	TC	9873	9916	10172	10223	10559	10373	10376	10378	10391	10571	10731	10735	10738	10738	10755	11410	11407	11415	11417	11423
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.68	0.68	0.68	0.68	0.68	0.69	0.69	0.69	0.69	0.69
35°F (1.7°C)	TC	10977	10978	11042	11073	11430	11490	11491	11491	11522	11850	11849	11850	11850	11851	12517	12514	12514	12513	12514	
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.70	0.70	0.71	0.71	0.74	0.75	0.75	0.75	0.75	0.75	0.78	0.78	0.78	0.78	0.78	0.85	0.85	0.85	0.85	0.85
41°F (5.0°C)	TC	10135	10135	10216	10287	10591	10606	10606	10607	10607	10644	10937	10937	10937	10937	10938	11551	11548	11547	11547	11548
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.58	0.58	0.58	0.59	0.61	0.61	0.61	0.61	0.61	0.62	0.64	0.64	0.64	0.64	0.64	0.70	0.70	0.70	0.70	0.70
47°F (8.3°C)	TC	9292	9298	9423	9465	9745	9723	9722	9723	9722	9768	10025	10024	10024	10024	10024	10586	10583	10582	10582	10583
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.45	0.46	0.46	0.47	0.49	0.48	0.48	0.48	0.48	0.49	0.51	0.51	0.51	0.51	0.51	0.55	0.55	0.55	0.55	0.55
50°F (10.0°C)	TC	9159	9411	9729	9785	10164	9676	9681	9748	9779	10174	10053	10066	10072	10072	10169	10746	10775	10802	10806	10819
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
59°F (15.0°C)	TC	9258	9583	9923	9983	10390	9796	9809	9920	9994	10400	10200	10210	10213	10228	10408	10952	10974	10996	10999	11008
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
68°F (20.0°C)	TC	9352	9751	10113	10177	10609	9901	9923	10124	10188	10620	10329	10337	10355	10364	10628	11129	11150	11168	11169	11175
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.57	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.59	0.59	0.59	0.59	0.59
77°F (25.0°C)	TC	9438	9892	10279	10348	10810	10005	10037	10291	10359	10581	10458	10470	10497	10509	10830	11296	11330	11356	11359	11369
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.61	0.61	0.61	0.61	0.62	0.61	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.61	0.62	0.62	0.62	0.62	0.62	0.62
82°F (27.8°C)	TC	9623	10114	10522	10593	11078	10197	10236	10531	10605	11090	10671	10685	10718	10731	11100	11538	11581	11612	11615	11627
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.67	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.69	0.69	0.69	0.69	0.69
86°F (30.0°C)	TC	9978	10476	10905	10981	11492	10523	10571	10917	10993	11504	11022	11034	11075	11091	11514	11932	11978	12007	12010	12021
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.78	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.80	0.80	0.80	0.80	0.80
95°F (35.0°C)	TC	10707	10765	11112	11178	11620	11379	11386	11389	11403	11635	11867	11870	11877	11878	11898	12788	12783	12785	12789	12806
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	0.99	0.99	1.01	1.01	1.03	1.02	1.02	1.02	1.02	1.03	1.04	1.04	1.04	1.04	1.04	1.08	1.08	1.08	1.08	1.08
104°F (40.0°C)	TC	9974	10065	10444	10507	10926	10602	10608	10627	10638	10939	11054	11059	11065	11065	11097	11915	11910	11916	11920	11933
	S/T	0.85	0.98	1.00	1.00	1.00	0.68	0.80	0.90	0.92	1.00	0.56	0.68	0.78	0.80	0.91	0.39	0.48	0.58	0.59	0.70
	PI	1.03	1.04	1.06	1.06	1.08	1.06	1.06	1.07	1.07	1.08	1.09	1.09	1.09	1.09	1.09	1.13	1.13	1.13	1.13	1.13
110°F (43.0°C)	TC	9497	9862	10238	10305	10751	10111	10114	10227	10317	10762	10566	10575	10577	10585	10772	11437	11437	11455	11457	11464
	S/T	0.92	1.00	1.00	1.00	1.00	0.72	0.87	0.99	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	1.07	1.09	1.10	1.10	1.12	1.10	1.10	1.10	1.10	1.12	1.12	1.12	1.12	1.12	1.12	1.15	1.15	1.15	1.15	1.15
115°F (46.0°C)	TC	8359	8762	9108	9176	9580	8890	8907	9119	9188	9590	9304	9310	9323	9332	9598	10088	10096	10110	10111	10114
	S/T	0.96	1.00	1.00	1.00	1.00	0.75	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.90	1.00	0.40	0.52	0.63	0.65	0.78
	PI	0.99	1.00	1.02	1.02	1.03	1.01	1.01	1												

Bolded values were tested. All other values are simulated and for reference only.

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9K Btu Single Zone Regular System - Ducted IDU



Heating Performance - Indoor Dry Bulb Temperature at 324 CFM										
Outdoor DB(°F)	TCI Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	5865	5747	5719	5696	5644	0.95	0.96	0.96	0.96	0.96
-4°F(-20.0°C)	7075	6934	6899	6871	6810	1.03	1.03	1.04	1.04	1.04
0°F(-17.8°C)	7851	7694	7656	7625	7557	1.06	1.07	1.07	1.08	1.08
5°F(-15.0°C)	8821	8645	8602	8567	8490	1.11	1.12	1.12	1.12	1.13
14°F(-10.0°C)	9700	9506	9459	9421	9336	1.13	1.14	1.14	1.14	1.15
17°F(-8.3°C)	9994	9794	9745	9706	9618	1.13	1.14	1.15	1.15	1.15
22°F(-5.6°C)	10323	10117	10066	10026	9935	1.14	1.15	1.15	1.15	1.16
27°F(-2.8°C)	10653	10439	10387	10346	10253	1.15	1.16	1.16	1.16	1.17
32°F(-0.0°C)	10982	10762	10709	10666	10570	1.15	1.16	1.16	1.17	1.17
35°F(1.7°C)	11180	10956	10901	10858	10760	1.16	1.17	1.17	1.17	1.18
37°F(2.8°C)	11713	11478	11421	11375	11273	1.16	1.17	1.17	1.17	1.18
42°F(5.6°C)	13045	12784	12721	12670	12556	1.17	1.18	1.18	1.19	1.19
44.6°F(7.0°C)	13578	13307	13240	13187	13069	1.18	1.19	1.19	1.19	1.19
47°F(8.3°C)	14378	14090	14020	13964	13838	1.18	1.19	1.19	1.20	1.20
52°F(11.1°C)	14185	13901	13832	13776	13652	1.08	1.09	1.09	1.09	1.10
57°F(13.9°C)	13992	13712	13643	13589	13467	0.98	0.98	0.99	0.99	0.99
59°F(15.0°C)	13915	13636	13568	13514	13392	0.93	0.94	0.94	0.95	0.95
62°F(16.7°C)	13799	13523	13455	13401	13281	0.87	0.88	0.88	0.88	0.89
64.4°F(18.0°C)	13706	13432	13365	13311	13192	0.82	0.83	0.83	0.83	0.84
69.8°F(21.0°C)	13472	13202	13136	13084	12966	0.71	0.72	0.72	0.72	0.72
75.2°F(24.0°C)	10035	9835	9786	9746	9659	0.60	0.60	0.61	0.61	0.61

Bolded values were tested. All other values are simulated and for reference only.